

EDUCATION**Univeristy of Wisconsin–Madison**

2023 - Present

Ph.D. in Physics

Advisor: Justin Vandenbroucke

Old Dominion University

2018-2023

B.S. Astrophysics and Computer Engineering

GPA: 3.80 (Overall), 3.83 (Major)

RESEARCH**Search for Neutrinos Coincident with the KM3NeT Neutrino**

March 2025 - Present

The IceCube Collaboration

Madison, WI

- PI: Professor Justin Vandenbroucke (UW)
- Searches for IceCube neutrinos coincident with KM3-230213A neutrino

Search for Neutrinos Coincident with Fast Radio Bursts

July 2024 - Present

The IceCube Collaboration

Madison, WI

- PI: Professor Justin Vandenbroucke (UW)
- Searches for neutrinos coincident with Fast Radio Bursts (FRBs) using CHIME's public Catalog-1
- Identification of potential particle acceleration mechanisms at FRBs as new neutrino sources

IceCube Follow-up of SN 2023ixf

August 2023 - July 2025

The IceCube Collaboration

Madison, WI

- PI: Professor Justin Vandenbroucke (UW)
- Neutrino Sources Working Group
- Performed IceCube follow-up of SN2023ixf, the closest and brightest supernova in nearly a decade

Missing Mass Group

January 2021 - May 2023

Old Dominion University

Norfolk, VA

- PI: Professor Balsa Terzic (ODU)
- Developed automated web scraping tool for CalTech's NASA/IPAC Extragalactic Database (NED)
- Developed nonlinear models to explore the consequences of self-interaction in spiral galaxies

Electrons for Neutrinos

February 2020 - August 2023

Old Dominion University/Thomas Jefferson National Accelerator Facility

Norfolk, VA

- PI: Dr. Lawrence Weinstein (ODU)
- Performed analysis on TJNAF electron beam data using CERN's ROOT library
- Used electron-neutron interactions to improve phenomenological models of neutrino interactions for modern experiments

INVITED TALKS**Nevada Center for Astrophysics Symposium**

April 2026

IceCube Follow-up of the KM3NeT Ultra-High-Energy Event[Invited Talk](#)**POSTERS AND TALKS****APS Global Summit - April Meeting**

March 2026

Results from IceCube Follow-up of the KM3NeT Ultra-high-energy Neutrino

Talk

APS Global Summit - March Meeting

March 2026

The Distributed Electronic Cosmic-ray Observatory

Talk

IceCube Fall 2025 Collaboration Meeting

October 2025

Results from IceCube Follow-ups of the KM3NeT Ultra-high Energy Neutrino

Plenary

30th Annual IPPOG Meeting <i>The Distributed Electronic Cosmic-ray Observatory</i>	October 2025 Talk
39th Meeting of the International Cosmic Ray Conference <i>Results from IceCube Follow-up of SN2023 ixf</i>	July 2025 Talk
UW-Madison Day at the Capitol <i>The IceCube Neutrino Observatory and Its Upgrades</i>	April 2025 Poster
244th Meeting of the American Astronomical Society <i>IceCube Follow-up of SN2023 ixf</i>	June 2024 Talk
2023 APS March Meeting <i>Probing NASA's Extragalactic Database for the Missing Mass Profile</i>	March 2023 Poster
Fall Meeting of the APS Division of Nuclear Physics <i>Electrons for Neutrinos: Analysis of the $1p1\pi$ Channel</i>	October 2021, 2022 Talk
Undergraduate Physics Congressional (PhysCon) <i>Rocket Propulsion, Probing NASA's Extragalactic Database for the Missing Mass Profile</i>	October 2019, 2022 Poster
Conference for Undergraduate Women in Physics (CuWiP) <i>Electrons for Neutrinos: Lepton Energy Reconstruction in the Resonance Excitation Region</i>	January 2021 Talk

SELECT PUBLICATIONS AND PROCEEDINGS

Publications where I had a major contribution in the work.

Results from IceCube Follow-up of Nearby Supernova SN2023ixf <i>Alicia Mand, Justin Vandenbroucke, Jessie Thwaites, Sam Hori, Aswathi Balagopal V</i>	September 2025 Proceedings of Science
The Distributed Electronic Cosmic-ray Observatory (DECO) <i>Justin Vandenbroucke, Alicia Mand, Runze Li, Alex Pizzuto</i>	September 2025 Proceedings of Science
On the Evidence for Violation of the Equivalence Principle in Disk Galaxies <i>Corey Sargent, William Clark, Antonia Seifert, Alicia Mand, et. al</i>	June 2025 Particles

TEACHING EXPERIENCE

Teaching Assistant <i>University of Wisconsin–Madison</i>	August - December 2023 <i>Madison, WI</i>
· Teaching Assistant for undergraduate algebra-based electricity and magnetism course (PHYS 104) · Graded assignments, held office hours, ran labs, and guided discussion sections	
Private Tutor at UW–Madison <i>University of Wisconsin–Madison</i>	August 2023 - Present <i>Madison, WI</i>
· Private tutor for undergraduate physics classes at UW Madison · Tutored students in algebra-based and calculus-based mechanics and E&M courses	
Undergraduate Teaching Fellow <i>Old Dominion University</i>	August 2022 - May 2023 <i>Norfolk, VA</i>
· Graded assignments and held office hours for introductory astronomy courses	

WORK EXPERIENCE

Research Assistant <i>University of Wisconsin–Madison</i>	January 2024 - Present <i>Madison, WI</i>
· Performed research under direction of Professor Justin Vandenbroucke for the IceCube Collaboration	

AWARDS

Wisconsin Space Grant Consortium Research Fellowship <i>Wisconsin Space Grant Consortium</i>	March 2026 \$6000
2025-2026 Wonders of Physics Outreach Fellow <i>University of Wisconsin–Madison</i>	September 2025 Outreach Fellowship
Gerald W. & Tui G. Hedstrom Award <i>University of Wisconsin–Madison</i>	April 2025 Departmental Award
2024-2025 Wonders of Physics Outreach Fellow <i>University of Wisconsin–Madison</i>	September 2024 Outreach Fellowship
2023-2024 Wonders of Physics Outreach Fellow <i>University of Wisconsin–Madison</i>	September 2023 Outreach Fellowship
Kaufman Award <i>Old Dominion University</i>	2023 2nd Place
ODU Department of Physics Outstanding Senior <i>Old Dominion University Department of Physics</i>	2023 \$250
APS Future of Physics Days Travel Award <i>American Physical Society</i>	2023 \$500
ODU SEES Travel Grant <i>Old Dominion University</i>	2023 \$500
SPS Travel Award <i>Society of Physics Students</i>	2023 \$300
Conference Experience for Undergraduates Travel and Lodging Award <i>Conference Experience for Undergraduates</i>	2022 \$550
JSA Minority/Female Undergraduate Research Assistantship <i>Thomas Jefferson National Laboratory</i>	2022 \$7,500
Virginia Space Grant Consortium Scholarship <i>Virginia Space Grant Consortium</i>	2022 \$8,500
College of Sciences General Scholarship <i>Old Dominion University</i>	2021 \$2,099
National Research Grant <i>Society of Physics Students</i>	2021 \$2,000
Pathway Retention Award <i>Old Dominion University</i>	2019-2021 Full Meal-Plan Waiver with Book Stipend

OUTREACH AND SERVICE

IceCube Summer School <i>Organizer</i> · Organizer for Summer 2026 IceCube Summer School	April 2026 - Present <i>Madison, WI</i>
Badgers on Tap <i>Speaker</i> · Public outreach talk on IceCube and multimessenger astronomy	April 2026 <i>Madison, WI</i>
APS TV Interview <i>Interview</i> · APS TV Interview on IceCube Follow-up of KM3NeT Ultra-high-energy neutrino	March 2026 <i>Denver, CO</i>
The Library Bar and Cafe Science Fair Night <i>Speaker</i> · Public outreach talk on IceCube and multimessenger astronomy	October 2025 <i>Madison, WI</i>

- Fundraising event for Madison Metropolitan School District Planetarium

Madison Metropolitan School District Planetarium

February 2025-December 2025

Volunteer

Madison, WI

- Collimated telescopes for telescope loan program
- Wrote show for annual Earth Day program on impacts of global climate change on amateur astronomy

University of Wisconsin–Madison Outreach Committee

August 2024-August 2025

Member

Madison, WI

- Member of the UW–Madison's Physics Department Outreach committee

Physics Graduate Student Council

August 2024 - Present

President

Madison, WI

- Active in recruitment and well-being of graduate student body
- Participated in recruitment events for prospective student visit weekends

The Wonders of Physics

February 2024, 2025

Performer

Madison, WI

- Wrote and performed fast-paced physics demos in 10 shows to a total audience of over 6,000 people of all ages

IceCube After School

January - February, 2024

Mentor

Madison, WI

- Led group of 3 high school students in the software development and implementation of LED IceCube model using Arduino

ADDITIONAL INFORMATION

Programming Languages

- Expert: Python, C++, Java, Arduino
- Proficient: MATLAB, C#, Assembly (x86 architecture), VHDL
- Skills: Computer Vision, Machine Learning, Parallel Programming, Networking, and Data visualization